

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN PHYSIOGRAPHY

MODULE - 8

PHYSIOGRAPHIC DIVISIONS **THE COASTAL PLAINS**

INDIA



MAJOR PHYSIOGRAPHIC DIVISION OF INDIA

1. The Northern and Northeastern Mountains
2. The Northern Plains
3. The Thar Desert
4. The Peninsular Plateau
5. The Coastal Plains
6. The Islands

THE COASTAL PLAINS



- 1 On the basis of the **location and active geomorphological processes**, the plains can be broadly divided into two:
 - The western coastal plains
 - The eastern coastal plains.
- 2 The western coastal plains are mostly an example of **submerged coastal plain**.
- 3 It is believed that the city of Dwaraka which was once a part of the Indian mainland situated along the west coast is submerged under water.
- 4 Because of this submergence **it is a narrow belt and provides natural conditions for the development of ports and harbours.**

INDIAN PHYSIOGRAPHY

THE COASTAL PLAINS



5 Kandla, Mazagaon, JLN port Nhava Sheva, Marmagao, Mangalore, Cochin, etc. are some of the important natural ports located along the west coast.

6 The western coastal plains are narrow in the middle and get broader towards north and south.

INDIAN PHYSIOGRAPHY

THE COASTAL PLAINS

7 The Malabar coast has got certain distinguishing features in the form of 'Kayals' (backwaters), which are used for fishing, inland navigation and also due to its special attraction for tourists.



8 Every year the famous Nehru Trophy Vallamkali (boat race) is held in Punnamada Kayal in Kerala.



THE COASTAL PLAINS

- 9** As compared to the western coastal plain, the eastern coastal plain is broader and is an example of an emergent coast.



- 10** There are well developed deltas here, formed by the rivers flowing eastward in to the Bay of Bengal.
- 11** These include the deltas of the Mahanadi, the Godavari, the Krishna and the Kaveri.
- 12** Because of its emergent nature, it has less number of ports and harbours.
- 13** The continental shelf extends up to 500 km into the sea, which makes it difficult for the development of good ports and harbours.

INDIAN PHYSIOGRAPHY

THE COASTAL PLAINS

WHY ARE THERE FEWER NATURAL PORTS IN THE EASTERN COAST OF INDIA?



- 1 West Rivers form estuaries and eastern rivers form delta. Therefore it becomes difficult for ships to reach the east coast covered with sediments.
- 2 West coast is coast of submergence (except Malabar Coast) while east coast is an emergent coast. These imply that sea is deeper in west coast than sea on east coast. So, west coast has favourable conditions for natural harbours.

THE COASTAL PLAINS

EAST COAST PLAIN

- 1 The East Coast plain lies along the East coast of India and is washed by the Bay of Bengal
- 2 The East Coast plain runs as a continuous broad plain
- 3 It is broader with width varying from 80 to 100 km
- 4 The large rivers make wide deltas on the East Coast

WEST COAST PLAIN

- 1 The West Coast plain lies along the West coast of India and is washed by Arabian Sea
- 2 The West Coast Plain is narrow and intersected by the mountain ranges
- 3 It is narrow with a width of 50 to 65 kms
- 4 The short swift rivers do not make any delta on the West Coast.

THE COASTAL PLAINS

EAST COAST PLAIN

- 5 The Eastern Coast is sandy with alluvium and slopes gently towards the sea. Sand dunes and marshy lands are also found, lagoons (Chilika, Pulicat) are formed
- 6 The Eastern coast is smooth, having large sediment deposits and therefore unfit for making ports.
- 7 This coastal plain receives relatively low rainfall.

WEST COAST PLAIN

- 5 The Western Coast is relatively rocky. It slopes abruptly down to the sea. It has many estuaries and back water with a series of lagoons.
- 6 The west coast is broken and indented and suitable for ports.
- 7 This coastal plain receives heavy rainfall.



1 The narrow coastal strip between the edges of the Peninsular Plateau and the coastline of India running for a distance of about six thousand kilometers from the Rann of Kuchchh in the west to the Ganga Brahmaputra delta in the east is called the ***Coastal Plains of India***.

2 The area between the Western Ghats and the Arabian sea coast is known as the ***West Coastal Plains*** and that between the Eastern Ghats and the Coast of the Bay of Bengal is called the ***East Coastal Plain***.

3 The two coastal plains meet each other at the southernmost tip i.e Kanyakumari

4

THE COASTAL PLAINS

1

THE WEST COASTAL PLAINS



1

Stretching from Rann of Kutch in the north to Kanyakumari in the South, there are narrow west coastal plains with an average width of about 65 km.

2

It is quite narrow in the middle and a bit broader in the northern and southern parts.

3

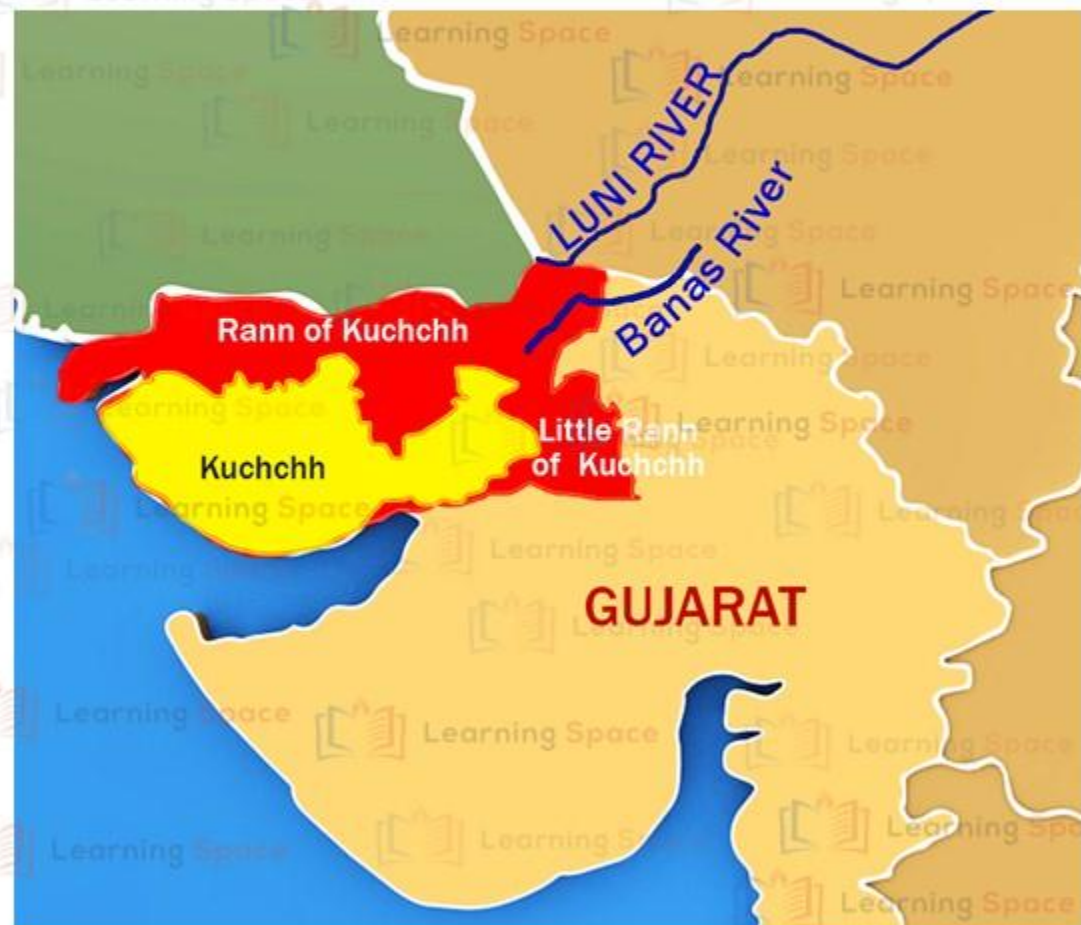
Depending upon relief and structure, it can be divided into following subdivisions.

1

THE WEST COASTAL PLAINS

A

THE KUCHCHH PENINSULA



1

The Kutchh Peninsula was an island surrounded by seas and lagoons.

2

These seas and lagoons were later filled by sediment brought by the Indus river which used to flow through this area.

3

Thus, the island became a part of the mainland and comparatively broad plain was formed.

4

Some scholars do not consider it as part of the west coastal plains and treat it as a separate identity.



Rann of Kuchchh

Kuchchh

Little Rann
of Kuchchh

GUJARAT

MADHYA PRADESH

MAHARASHTRA

1

THE WEST COASTAL PLAINS

A

THE KUCHCHH PENINSULA

**5**

There are other geographers who consider **Kachchh and Kathiawar as part of the Peninsular Plateau** because Kathiawar is made of the Deccan Lava and there are tertiary rocks in the Kuchchh area.

6

The true west coastal plain, according to them, **lies between Surat and Kanniyakumari** for a distance of 1,600 Km.

7

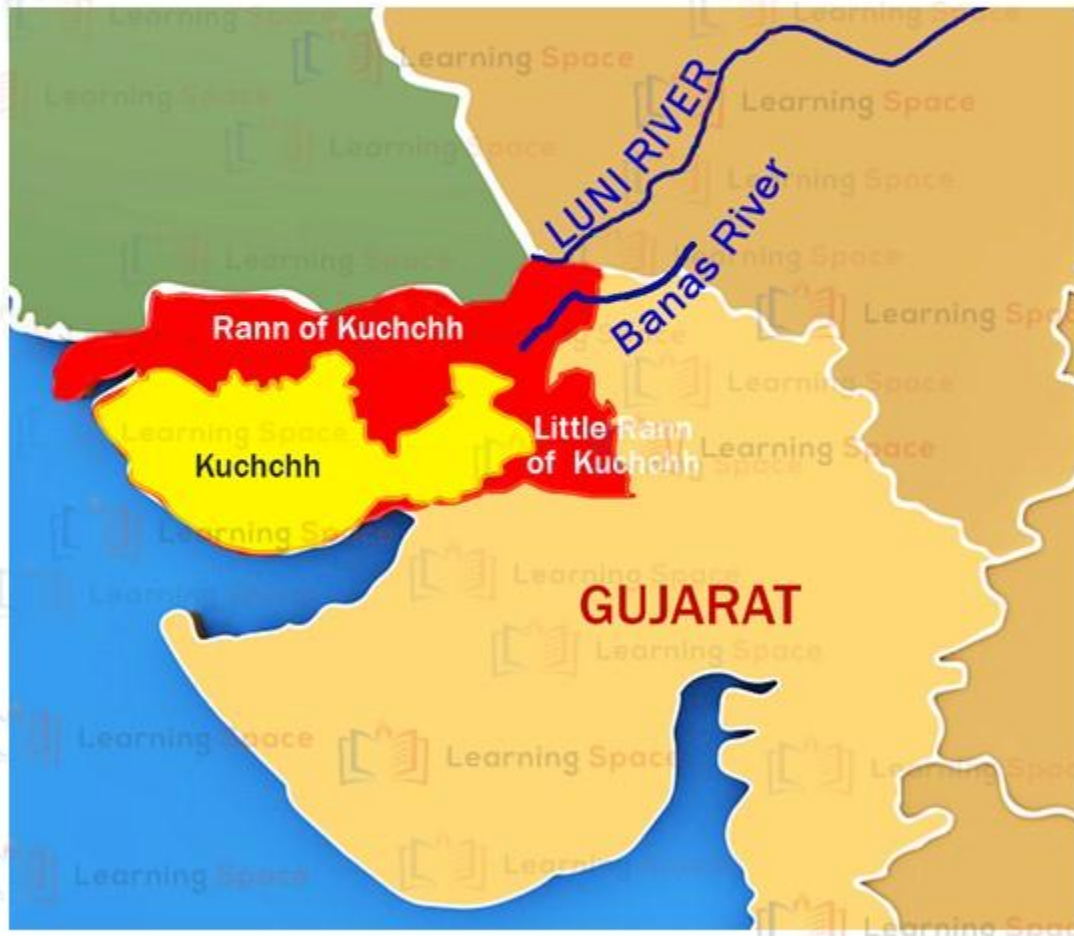
But the ground reality is **that it is, more or less, a plain area and lies near the west coast of India**. Hence it should be treated as an integral part of the West Coastal Plains.

1

THE WEST COASTAL PLAINS

A

THE KUCHCHH PENINSULA

**8**

Due to scarcity of rain and flowing surface water, the work of wind is felt everywhere and this has given rise to arid and semi-arid landscape.

9

Coastal sand dunes, sandy plains, interrupted with bare rocky hills are the chief characteristic physiographic features.

10

All along the north of Kutchch there lies a broad level salt-soaked plain. This is the Great Rann.

11

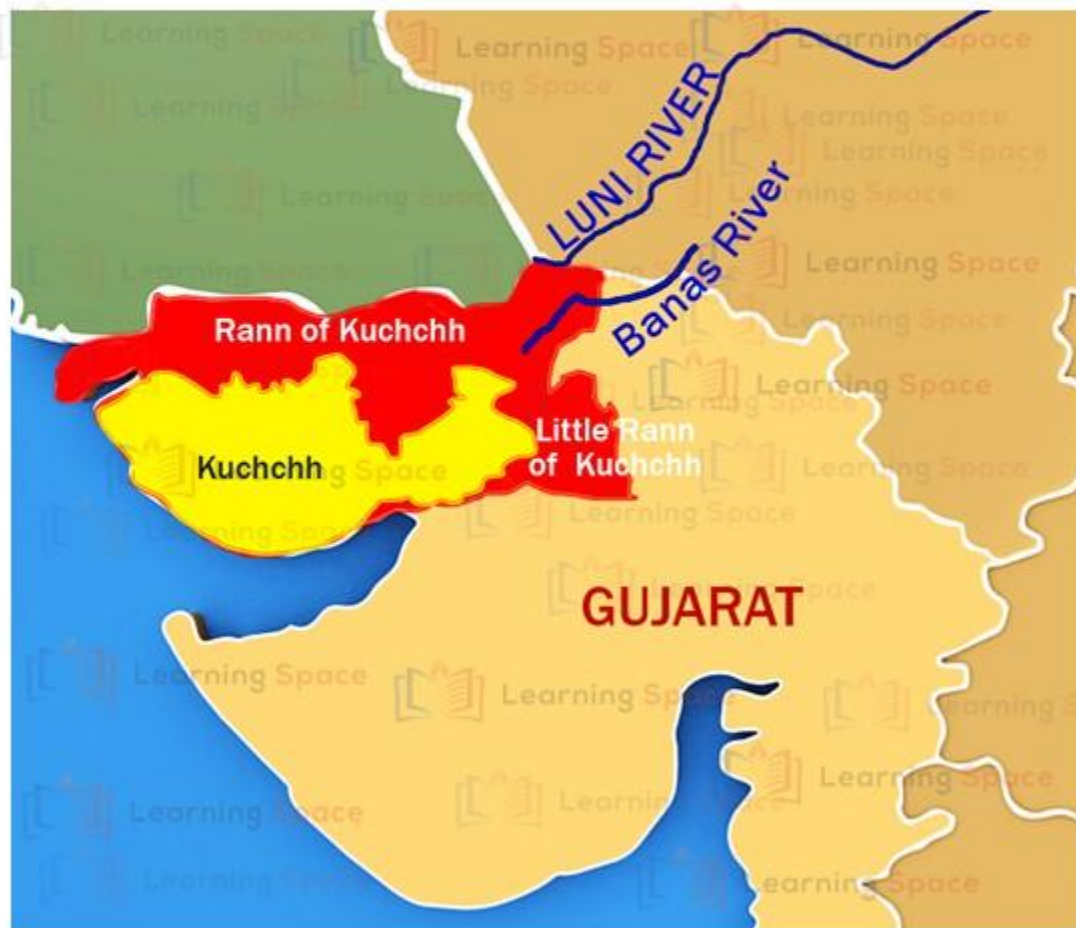
Its southern continuation, known as the Little Rann lies on the south-east of Kutchch.

1

THE WEST COASTAL PLAINS

A

THE KUCHCHH PENINSULA

**12**

The flat and unbroken Great Rann is about 320 kms long with a maximum width of 60 km, covering an area of about 21,500 sq km.

13

It rises only a few metres above the sea level and is flooded by the Banas and the Luni rivers during the rainy season.

14

Some places are actually below sea level and are inundated during high tides.

15

Most of the area is formed of sun-baked dark silt encrusted with salt. A few patches of high ground are covered with grass and break the monotony of the otherwise flat plain.

1

THE WEST COASTAL PLAINS

B

THE KATHIAWAR PENINSULA



1

The Kathiawar Peninsula lies to the south of the Kutchchh.

2

It is encircled on the east and north-east by the Little Rann and the Nal Basin.

3

The average elevation is less than 200 metres.

4

The central part is a highland of Mandav hills from which small streams radiate in all directions.

5

Mt. Girnar (1,117 m) the highest point, is supposed to be of volcanic origin.

6

The Gir range is located in the southern part of the Kathiawar Peninsula. It is covered with dense forests and is famous as the home of Gir lion.

Kuchchh Peninsula

Mandav Hills

Kathiawar Peninsula

Gir Range



1

THE WEST COASTAL PLAINS

C

THE GUJARAT PLAIN

1 The Gujarat Plain lying east of Kutchh and Kathiawar slopes towards the west and south west.

2 It may almost be described as an intrusion of Indo-Gangetic conditions into the Peninsula.

3 Formed by the rivers Narmada, Tapi, Mahi and Sabarmati the plain includes the southern part of Gujarat and the coastal areas of Gulf of Khambhat. This is a low plain no part of which exceeds 150 m elevation.

4 The eastern part of this plain is made of sediments and is fertile enough to support agriculture, but the greater part near the coast is covered by windblown loess, which has given rise to semi-arid landscape.

5 A chain of saline marshes near the coast is prone to floods during high tide.



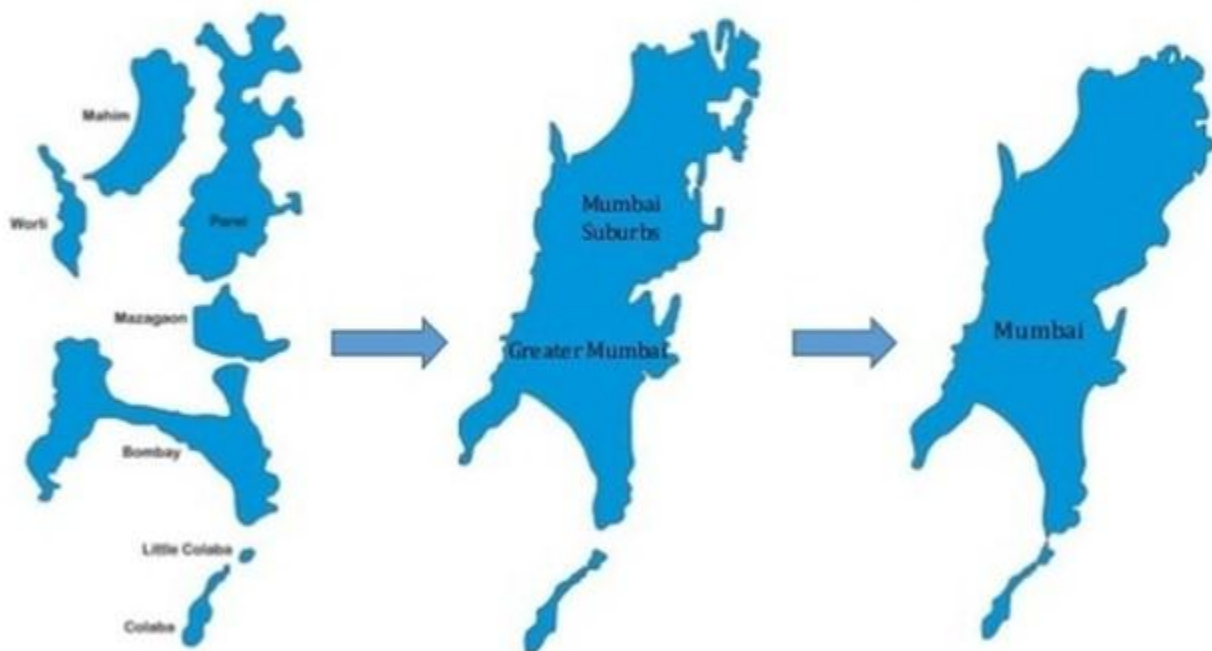
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THE WEST COASTAL PLAINS

D

THE KONKAN PLAIN

SEVEN ISLANDS TO MUMBAI CITY



Reclamation of land to form present day Mumbai

1

The Konkan Plain south of the Gujarat plain extends from **Daman to Goa** for a distance of about 500 km with its width varying from 50 to 80 kms.

2

It has **some features of marine erosion** including cliffs, shoals, reefs and islands in the Arabian sea.

3

Mumbai was an island, but parts of the sea lying between the mainland and island have been **reclaimed** in recent years to connect it with the mainland.

1

THE WEST COASTAL PLAINS

D

THE KONKAN PLAIN

**4**

The Thane creek of the Ulhas around Mumbai is an important embayment which provides an excellent natural harbor on the southern side of the island.

5

South of the Mumbai, the Konkan coast has a series of small bays and coves containing beaches of sand.

6

Behind the alluvial coastal belt, there is series of parallel ridges reaching 450-600 metres in which rivers like the Vaitarni, Ulhas and Amba have lower courses more or less parallel to the coast before reaching it transversely.

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